Mereology Hands-on

Or what happens when one tries to approximate a FOL theory in OWL with Protégé

Basic Mereology

- **Create a new ontology in Protégé**
- **No other class than Thing, three properties p, o and pp**
- **Try to capture the axioms, definitions and theorem:**
  - Reflexivity $\forall x\, P(x,x)$
  - Transitivity $\forall xyz\, ((P(x,y) \land P(y,z)) \rightarrow P(x,z))$
  - Antisymmetry $\forall xy\, ((P(x,y) \land P(y,x)) \rightarrow x=y)$
  - ProperPart $PP(x, y) \triangleq P(x, y) \land \neg x = y$
  - Overlap $O(x, y) \triangleq \exists z(P(z, x) \land P(z, y))$
  - Irreflexivity $\forall x\, \neg PP(x,x)$
- **Translation from FOL to OWL-2 in Protégé: is it all expressible?**
- **Can all be inserted in the ontology together?**
  Check: add instances + run reasoner

Translation

- Class inclusion: $C_1 \sqsubseteq C_2$ $\forall x\ (C_1(x) \rightarrow C_2(x))$
- Property chain: $R_1 \circ R_2 \sqsubseteq R_3$ $\forall xy\ (\exists z(R_1(x,z) \land R_2(z,y)) \rightarrow R_3(x,y))$
  NB: transitivity implemented as a feature in Protégé
- Existential restriction on a class:
  $\exists\ R.C_1 \sqsubseteq C_2$ $\forall x\ (\exists y(R(x,y) \land C_1(y)) \rightarrow C_2(x))$
  NB: (ir)reflexivity implemented as a feature in Protégé (exploiting T and Self construct: $\top \sqsubseteq \exists P.\mathit{Self}$ and $\top \sqsubseteq \neg\exists PP.\mathit{Self}$)
- Universal restriction on a class:
  $\forall\ R.C_1 \sqsubseteq C_2$ $\forall x\ (\forall y(R(x,y) \rightarrow C_1(y)) \rightarrow C_2(x))$

Not just a language expressivity issue

- ➢ Antisymmetry not expressible
- ➢ PP not definable, only subproperty of P
- ➢ Definition of O partially captured with property chain (one sense only)
- ➢ One can nevertheless add some theorems as axioms: P subproperty of O

- But additional constraints to guarantee decidability!
  Simplicity and Regularity
- A Description Logic Primer
  Markus Krötzsch, František Simancík, Ian Horrocks
  https://arxiv.org/pdf/1201.4089

- ➢ Transitivity and irreflexivity of PP cannot be adopted simultaneously!
- ➢ Basic mereology very partially implemented in OWL 🙁

Let’s add time (and change)!

- **Add a top-level:**
  Object, Events, Times as disjoint subclasses of Thing
- Non-temporalized p applies to Events and Times only
  - **Add domain and range restrictions for p, o, pp**
  - What happens?

Let’s add time (and change)!

- **Add a top-level:**
  Object, Events, Times as disjoint subclasses of Thing
- Non-temporalized p applies to Events and Times only
  - **Add domain and range restrictions for p, o, pp**
  - What happens?
  - Issue with reflexivity: restricting it to the domain breaks regularity!
- **Add temporalized P_t for Objects**
  - Transitivity $\forall xyzt\ ((P_t(x,y,t) \land P_t(y,z,t)) \to P_t(x,z,t))$
  - (forget about reflexivity) $\forall xt\ P_t(x,x,t)$
  - (forget about antisymmetry) $\forall xy\ (\forall t\ (P_t(x,y,t) \land P_t(y,x,t)) \to x=y)$

➢ What to do with ternary relations in OWL?

Two approaches

- Focus on constant parthood CP (no change)
  - Interaction with an « existence » predicate for endurants E(x,t)
- Go « neo-davidsonian » and use the W3C recommendation: reification
  - Defining N-ary Relations on the Semantic Web: Use With Individuals https://www.w3.org/2001/sw/BestPractices/OEP/n-aryRelations-20040623/
- One can do both!

- **How would you implement the reification of P_t?**

Full(er) mereology?

- Extensionality
  - Strong supplementation $\forall xy\, (\neg P(y, x) \rightarrow \exists z(P(z, y) \wedge \neg O(z, x)))$
- Composition principles
  - Sum definition $Sum(s, x, y) \triangleq \forall z(O(z, s) \leftrightarrow (O(z, x) \vee O(z, y)))$
  - Sum existence $\forall xy\, \exists s(Sum(s, x, y))$

Explore foundational ontologies

- **BFO, UFO, DOLCE**
  https://github.com/appliedontolab/DOLCE
- **Identify the mereological relations and their axiomatization**
  - Which relations?
  - Which axioms?
  - If any, what approach to temporalized parthood?
- Reference: Approximating DOLCE in OWL: The DOLCEbasic and DOLCEnaryRel Core Modules, Porello et al., *Applied Ontology*, to appear
  in the course material